

RaceDay System – API Endpoint Plan

PROG6212 Programming 2B | Part 1 – System Planning and Database

1. Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new RaceDay user and assigns the selected role. The password is hashed before storage.	None	{ "email": "user@example.com", "password": "Password123", "roleId": 1 }	201 Created – User registered successfully. 400 Bad Request – Invalid data. 409 Conflict – Email already exists.
POST	/api/auth/login	Authenticates a user and creates a session containing the user's ID and role.	None	{ "email": "user@example.com", "password": "Password123" }	200 OK – Login successful and session established. 401 Unauthorized – Invalid credentials.
POST	/api/auth/logout	Ends the authenticated user's session.	Any	None	200 OK – Logout successful. 401 Unauthorized – No active session.

2. User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Retrieves the profile of the currently authenticated user.	Any	None	200 OK – User profile. 401 Unauthorized – User is not logged in.
PUT	/api/users/me	Updates the authenticated user's profile information.	Any	{ "email": "newemail@example.com" }	200 OK – Updated profile. 400 Bad Request – Invalid data. 409 Conflict – Email already exists.
PUT	/api/users/me/password	Changes the authenticated user's password after validating the current password.	Any	{ "currentPassword": "...", "newPassword": "..." }	200 OK – Password updated. 400 Bad Request – Invalid password data. 401 Unauthorized – Current password incorrect.

3. Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Retrieves all available events for browsing.	None	None	200 OK – List of events.
GET	/api/events/{eventId}	Retrieves complete information for a specific event.	None	None	200 OK – Event details. 404 Not Found – Event does not exist.
POST	/api/events	Creates a new event owned by the authenticated Organiser.	Organiser	{ "name": "...", "description": "...", "date": "...", "location": "...", "distance": 21, "eventType": "Run" }	201 Created – Event created. 400 Bad Request – Invalid event data. 401 Unauthorized – Not authenticated. 403 Forbidden – Participant attempted access.
PUT	/api/events/{eventId}	Updates an event created by the authenticated Organiser.	Organiser	{ "name": "...", "description": "...", "date": "...", "location": "...", "distance": 21, "eventType": "Run" }	200 OK – Event updated. 404 Not Found – Event does not exist. 403 Forbidden – Organiser does not own event.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
DELETE	/api/events/{eventId}	Deletes an event created by the authenticated Organiser.	Organiser	None	204 No Content – Event deleted. 404 Not Found – Event does not exist. 403 Forbidden – Access denied.

4. Event Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Retrieves all categories available for an event.	None	None	200 OK – List of categories. 404 Not Found – Event does not exist.
GET	/api/events/{eventId}/categories/{categoryId}	Retrieves a specific category belonging to an event.	None	None	200 OK – Category details. 404 Not Found – Category does not exist.
POST	/api/events/{eventId}/categories	Creates an age or distance category for an event.	Organiser	{ "name": "Senior", "categoryType": "Age" }	201 Created – Category created. 400 Bad Request – Invalid data. 403 Forbidden – Access denied.
PUT	/api/events/{eventId}/categories/{categoryId}	Updates an existing event category.	Organiser	{ "name": "Senior", "categoryType": "Age" }	200 OK – Category updated. 404 Not Found – Category not found.
DELETE	/api/events/{eventId}/categories/{categoryId}	Removes a category from an event.	Organiser	None	204 No Content – Category deleted. 404 Not Found – Category not found.

5. Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{eventId}/enrolments	Enrols the authenticated Participant into an event using a selected category.	Participant	{ "categoryId": 1 }	201 Created – Enrolment created. 404 Not Found – Event/category not found. 409 Conflict – Participant already enrolled. 403 Forbidden – User is not a Participant.
GET	/api/enrolments/me	Retrieves all events the authenticated Participant has entered.	Participant	None	200 OK – Participant's enrolments.
GET	/api/events/{eventId}/enrolments	Retrieves all Participants enrolled in a specific event.	Organiser	None	200 OK – List of event enrolments. 403 Forbidden – Organiser does not own event. 404 Not Found – Event not found.
GET	/api/enrolments/{enrollmentId}	Retrieves details of a specific enrolment.	Any authenticated user	None	200 OK – Enrolment details. 403 Forbidden – User does not have access. 404 Not Found – Enrolment not found.

6. Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments/{enrollmentId}/result	Records a Participant's finishing time and position after an event.	Organiser	{ "finishTime": "01:45:32", "position": 47 }	201 Created – Result recorded. 404 Not Found – Enrolment not found. 409 Conflict – Result already exists.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/results/me	Retrieves the authenticated Participant's complete personal race history.	Participant	None	200 OK – List of personal results.
GET	/api/events/{eventId}/results	Retrieves all results for a completed event.	Organiser	None	200 OK – Event results. 404 Not Found – Event not found. 403 Forbidden – Access denied.
GET	/api/results/{resultId}	Retrieves a specific race result.	Any authenticated user	None	200 OK – Result details. 404 Not Found – Result not found.
PUT	/api/results/{resultId}	Updates an existing finish time or finishing position.	Organiser	{ "finishTime": "01:44:18", "position": 46 }	200 OK – Result updated. 404 Not Found – Result not found. 403 Forbidden – Access denied.